

UNIVERSITY OF BAHRAIN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER 2022 -2023

TEST 2 (B) - Key

MATHS. 211

DATE: 5 - 12 - 2022

TIME: 12:00 PM - 1:00 PM

EXAM INSTRUCTIONS:

- 1) Number of pages is 5 including the cover page.
 - 2) Show all your work, except for question one.
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NAME :	I.D. NUMBER:	SECTION:	SERIAL NUMBER:

Q.1	Q.2	Q.3	Q.4	OUT OF
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QUESTION 1 (2 marks for each question)

Fill in the blanks to make a correct statement:

1) Suppose that $(A)_S = (2, 1, -1, 2)$ where $S = \left\{ \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} \right\}$ is a basis of M_{22} , then

$$A = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}$$

2) If $S = \{p_1, p_2, p_3, p_4, p_5\}$ spans P_n , then

$$n \leq 4$$

3) If A, B are two nonzero matrices such that $B \in \text{Span}\{A\}$, then $\{A, B\}$ is

linearly dependent

4) If $A' = \begin{bmatrix} 1 & 2 & 5 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is a row-echelon form matrix of A , then a basis of the row space of A is

$$\{(1, 2, 5, 1), (0, 0, 1, 1)\}$$

5) If A is a matrix of size 4×7 , then the largest possible value of $\text{rank}(A)$ is equal to

$$4$$

6) If $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 2 & 2 & 3 \end{bmatrix}$, then the rank of A is

$$3$$

7) If A is a matrix of size 8×4 , and $\text{nullity}(A^T) = 3$, then

$$\text{nullity}(A) = 2$$

8) If $\{(t, 1, 1), (0, 1, 2), (0, 0, 1)\}$ is a basis of $\mathbb{R}^3$, then

$$t \neq 0$$

QUESTION 2 (4 + 4 + 2 marks)

(Show your work)

a) Show that $S = \{p_1, p_2, p_3\}$ is a basis of P_2 , where

$$p_1 = 2 + X, \quad p_2 = X + X^2, \quad p_3 = 2 + X^2$$

Solution: Since S has 3 elements and $\dim P_2 = 3$, it is sufficient to prove that S is linearly independent.

$$\begin{aligned} ap_1 + bp_2 + cp_3 = 0 &\Rightarrow a(2 + X) + b(X + X^2) + c(2 + X^2) = 0 \\ &\Rightarrow (2a + 2c) + (a + b)X + (b + c)X^2 = 0 \\ &\Rightarrow \begin{aligned} 2a + 2c &= 0 \\ a + b &= 0 \\ b + c &= 0 \end{aligned} \\ &\Rightarrow a = b = c = 0 \end{aligned}$$

b) Find the coordinates of $p = 1 + X + X^2$ relative to S .

Solution:

$$\begin{aligned} ap_1 + bp_2 + cp_3 = p &\Rightarrow a(2 + X) + b(X + X^2) + c(2 + X^2) = 1 + X + X^2 \\ &\Rightarrow (2a + 2c) + (a + b)X + (b + c)X^2 = 1 + X + X^2 \\ &\Rightarrow \begin{aligned} 2a + 2c &= 1 \\ a + b &= 1 \\ b + c &= 1 \end{aligned} \\ &\Rightarrow a = c = 1/4, \quad b = 3/4 \\ &\Rightarrow (p)_S = (1/4, 3/4, 1/4) \end{aligned}$$

c) Without any calculations, determine whether $\{p_1, p_2, p_3, p\}$ is linearly independent.

Solution:

The set $\{p_1, p_2, p_3, p\}$ is linearly dependent because it consists of 4 vectors in the vector space P_2 with dimension 3.

QUESTION 3 (6 marks)

Find the dimension of $V = \text{Span}\{f_1, f_2, f_3\}$, where $f_1 = x$, $f_2 = \cos(x)$, $f_3 = \sin(x)$.

Solution:

The set $\{f_1, f_2, f_3\}$ is a spanning set of V . To prove that it is a basis, it suffices to show that it is linearly independent, equivalently to prove that $W(x)$ is not the zero function.

We have

$$W(x) = \begin{vmatrix} x & \cos(x) & \sin(x) \\ 1 & -\sin(x) & \cos(x) \\ 0 & -\cos(x) & -\sin(x) \end{vmatrix}$$

By adding the row 1 to the row 3, we get

$$W(x) = \begin{vmatrix} x & \cos(x) & \sin(x) \\ 1 & -\sin(x) & \cos(x) \\ x & 0 & 0 \end{vmatrix}$$

So

$$W(x) = x \begin{vmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{vmatrix}$$

That is,

$$W(x) = x ((\cos(x))^2 + (\sin(x))^2) = x \neq \text{zero function}.$$

QUESTION 4 (4 + 4 marks)

Let $A' = \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ be a row-echelon form of A .

a) Find a basis of the nullspace of A .

Solution:

Let us transform the matrix A' to a reduced row-echelon form A'' :

$$A' \sim \begin{bmatrix} \mathbf{1} & 3 & 0 & 2 \\ 0 & \mathbf{1} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} \mathbf{1} & 3 & 0 & 0 \\ 0 & \mathbf{1} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} \mathbf{1} & 0 & -3 & 0 \\ 0 & \mathbf{1} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = A''$$

$$\begin{bmatrix} \mathbf{1} & 0 & -3 & 0 \\ 0 & \mathbf{1} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} X = O \Rightarrow x_1 - 3x_3 = 0 ; x_2 + x_3 = 0 ; x_4 = 0$$

Let $x_3 = r$ be a free variable, then the leading variables are $x_1 = 3r$, $x_2 = -r$ and $x_4 = 0$.

Thus $N(A) = \{ (3r, -r, r, 0) : r \text{ is real number} \} = \text{Span}\{(3, -1, 1, 0)\}$.

Since $\dim N(A) = \text{Nullity}(A) = 4 - 3 = 1$, then $\{(3, -1, 1, 0)\}$ is a basis of $N(A)$.

b) If c_1, c_2, c_3, c_4 are the column vectors of A , write c_3 as a linear combination of c_1, c_2, c_4 .

Solution:

Let c_1', c_2', c_3', c_4' be the column vectors of A'' .

Then $c_3' = -3c_1' + c_2'$. It follows that $c_3 = -3c_1 + c_2$